

STEPHANIE KRIE WALL

DATA SCIENCE | MACHINE LEARNING

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SUMMARY

Versatile data science professional with extensive programming and statistical analysis experience and a proven track record of adapting to new domains. Master's in Statistics with a strong ability to transform complex data into actionable insights that support data-driven decision making and problem solving.

SKILLS

Programming: Python, SQL, R, C++, SAS, JavaScript

Machine Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, k-means, PCA, Association Rule Learning, Causal Impact Analysis

Tools: Excel, Tableau, Git, AWS, Anaconda

Math: Statistics, A/B Testing, Time Series Analysis, Bayesian Statistics, Linear Algebra

PROJECTS

Enhancing Targeting Accuracy Using Machine Learning

- Used [logistic regression](#), [decision tree](#), [random forest](#), and [KNN](#) models on grocery transaction data to predict customer sign-up probabilities for a grocery delivery service based on available customer data, enabling better targeting decisions based on which customers are most likely to sign up.

"You Are What You Eat" Customer Segmentation

- Performed [k-means clustering](#) on grocery transaction data to split customers into distinct "shopper types" that could be used to more accurately target customers with relevant content and promotions and to better understand customers over time.

Additional portfolio projects at skriewall.github.io.

EXPERIENCE

Software Developer – Mentice

MAY 2019 – AUGUST 2025

- Developed and maintained core simulation framework for endovascular and neurovascular procedures (VIST product line), enabling hands-on training with real-time physics and medical device interaction. Used by 700+ customers – including hospitals, universities, simulation centers, and top 20 MedTech companies – to support clinical decision-making and improve patient outcomes.
- Created a 3D augmented reality surgical training tool in Unreal Engine, providing immersive visualization of heart procedures for enhanced clinician engagement and learning.
- Solved a critical Unreal Engine adaptation challenge for medical simulation by implementing the centripetal parameterization of Catmull-Rom splines, preventing cusps and self-intersections in tortuous vascular anatomy.
- Researched and developed an adaptive spline simplification algorithm to reduce control point density and vessel mesh complexity while preserving anatomical fidelity, significantly improving runtime performance in high-complexity cases.
- Automated build workflows using Python, including a custom installer that seamlessly integrates Unreal Engine applications with the VIST simulation environment.

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EXPERIENCE (CONT.)

Software Developer - Mentice (Cont.)

MAY 2019 – AUGUST 2025

- Contributed to the full development lifecycle as part of a cross-functional Agile team, collaborating closely with fellow engineers and 3D designers.

Mathematical Analyst - Gaming Laboratories International

MARCH 2015 – MAY 2019

- Developed C++ game simulations and used probability models to analyze the return to player and odds for slot machine games.
- Performed statistical analysis of simulations and conducted RNG randomness testing to ensure regulatory compliance.
- Evaluated player strategy and house edge for table and slot games using concepts including probability theory, expected value, Markov decision processes, dynamic programming, and game theory.
- Trained new analysts and conducted detailed technical reviews to ensure accuracy and consistency in game evaluations.

Analyst - FIMAC Solutions

NOVEMBER 2014 – MARCH 2015

- Conducted prepayment modeling and other statistical analysis to support risk and performance assessments for financial institutions.
- Developed code for integration into company software products.

EDUCATION

M.S. (Statistics)

2012 - 2014 – University of Iowa, Iowa City, IA

B.A. (Mathematics)

2006 - 2010 – Wisconsin Lutheran College, Milwaukee, WI